

FAKE NEWS DETECTION USING MACHINE LEARNING AND NLP



Team Members

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PROBLEM STATEMENT

The Problem

Misinformation and fake news spread rapidly across digital platforms, influencing public opinion and creating social, political, and economic harm at a massive scale.

Why It Matters

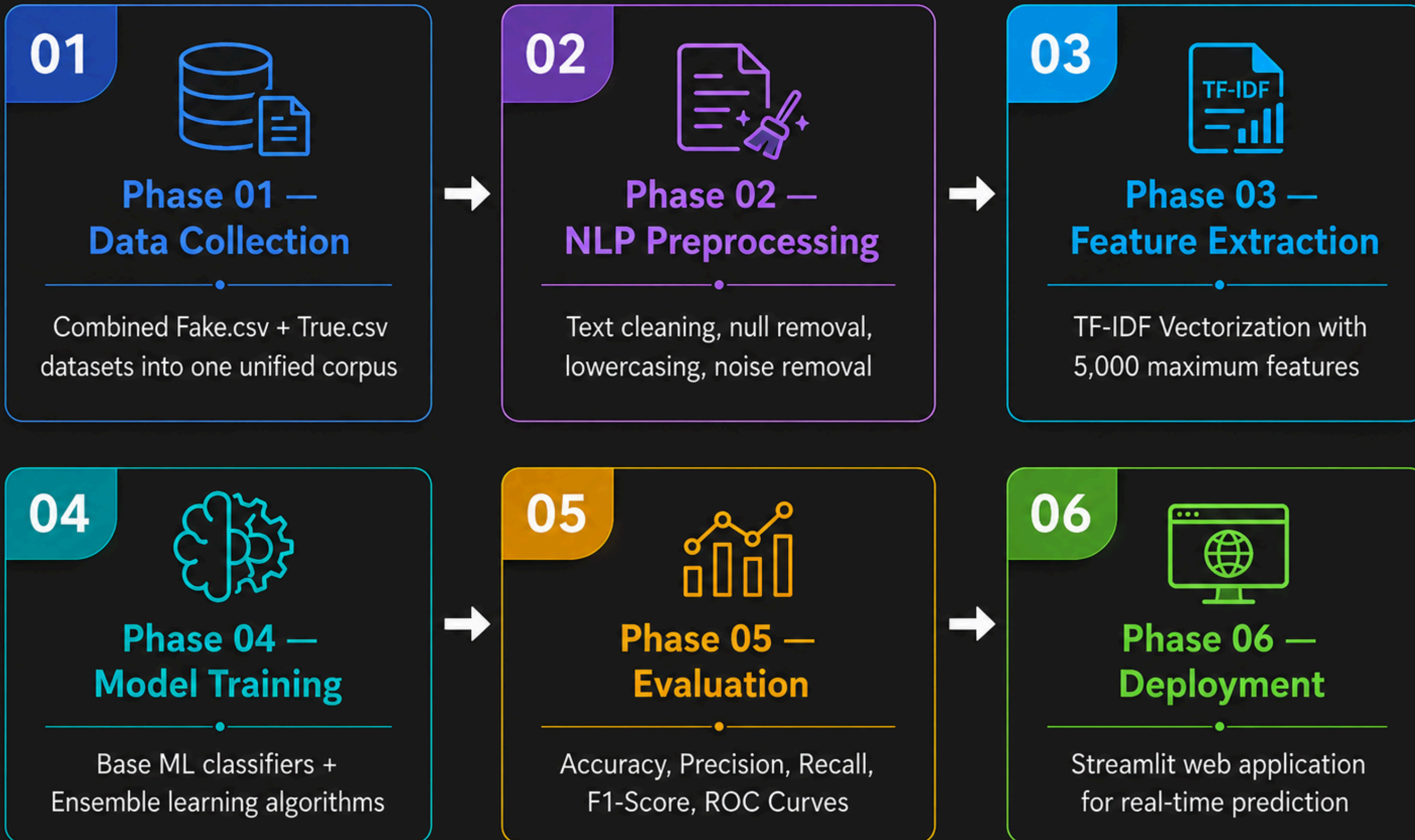
Manual verification of news is slow, inconsistent, and impossible to scale with the huge volume of online content. An automated system is needed for fast and reliable detection.

Our Objective

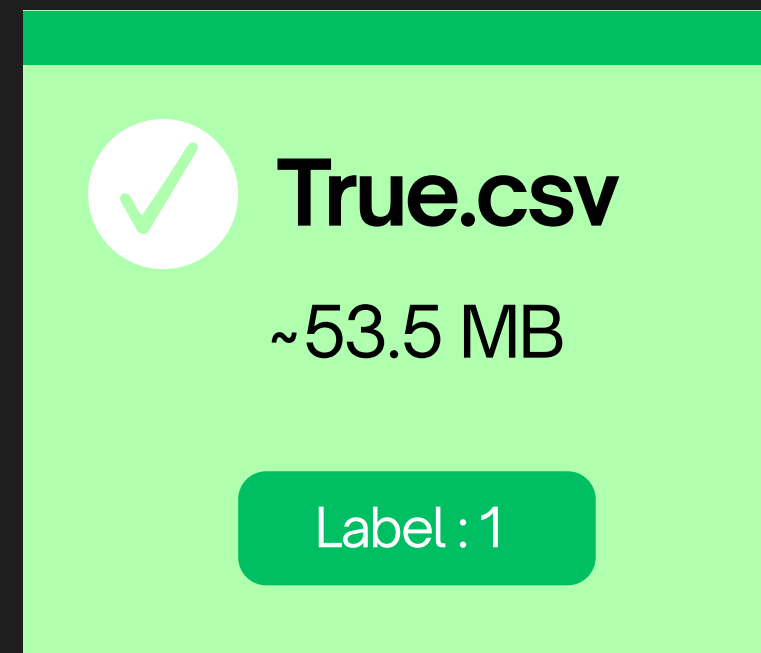
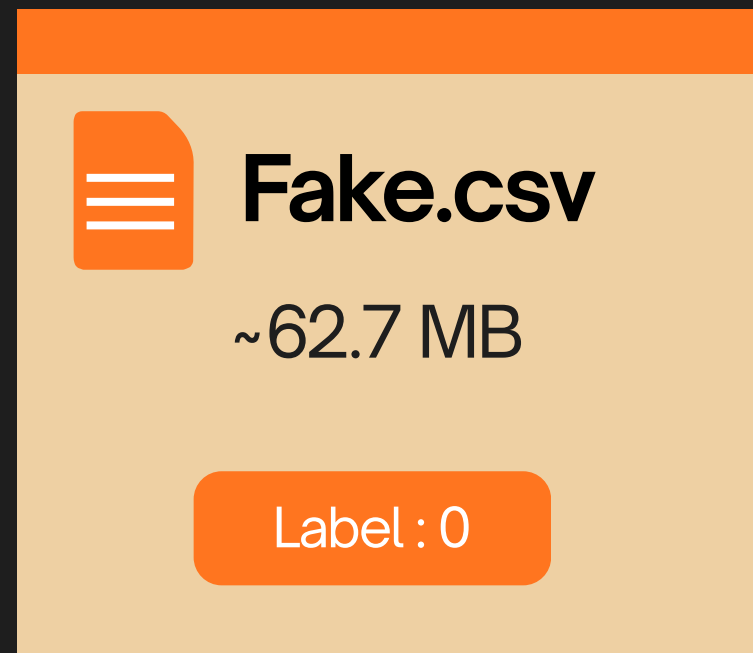
To develop and evaluate Machine Learning and Ensemble models that can accurately classify news articles as Real or Fake using only textual content.



METHODOLOGY & ARCHITECTURE

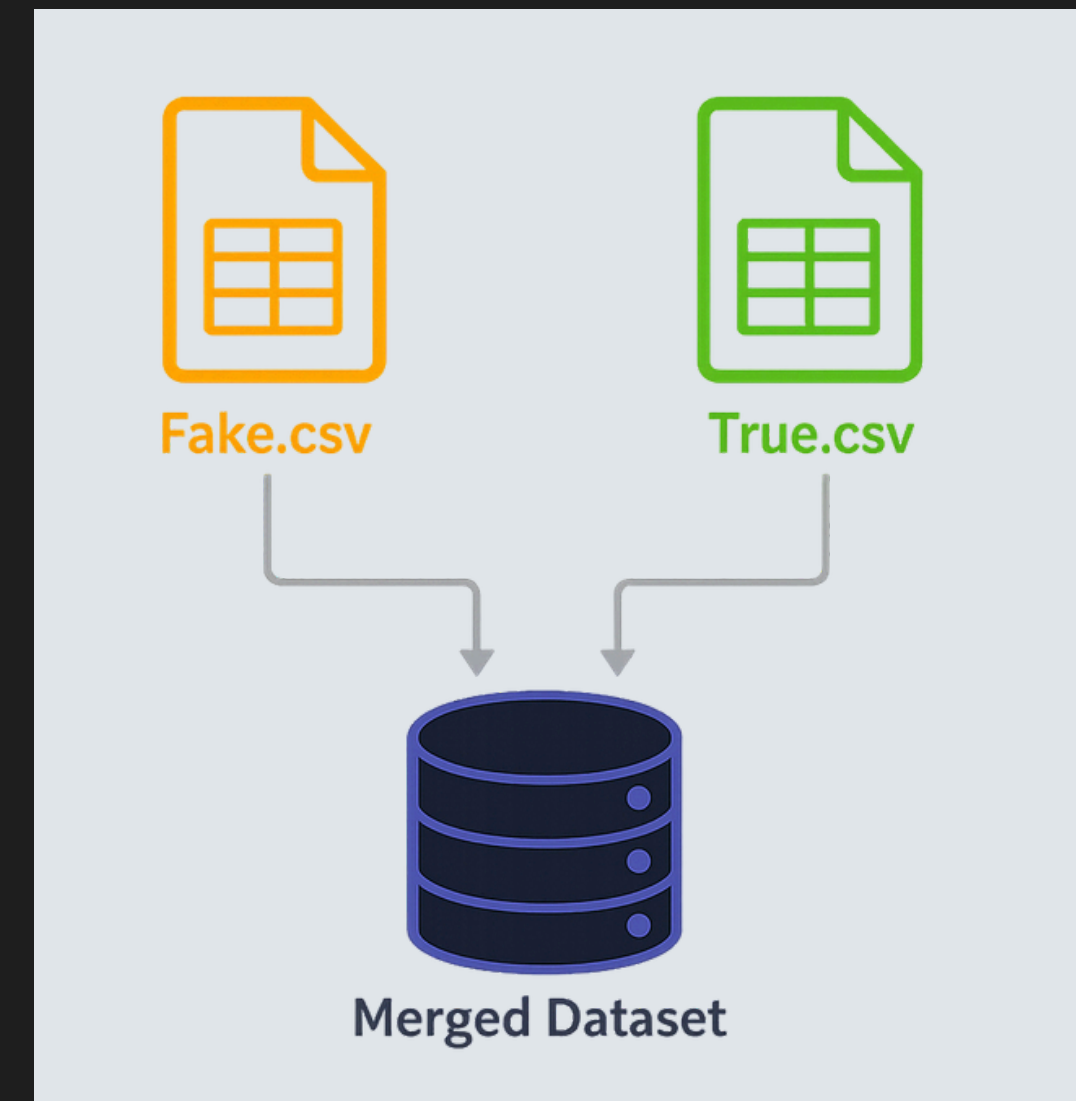


DATASET OVERVIEW

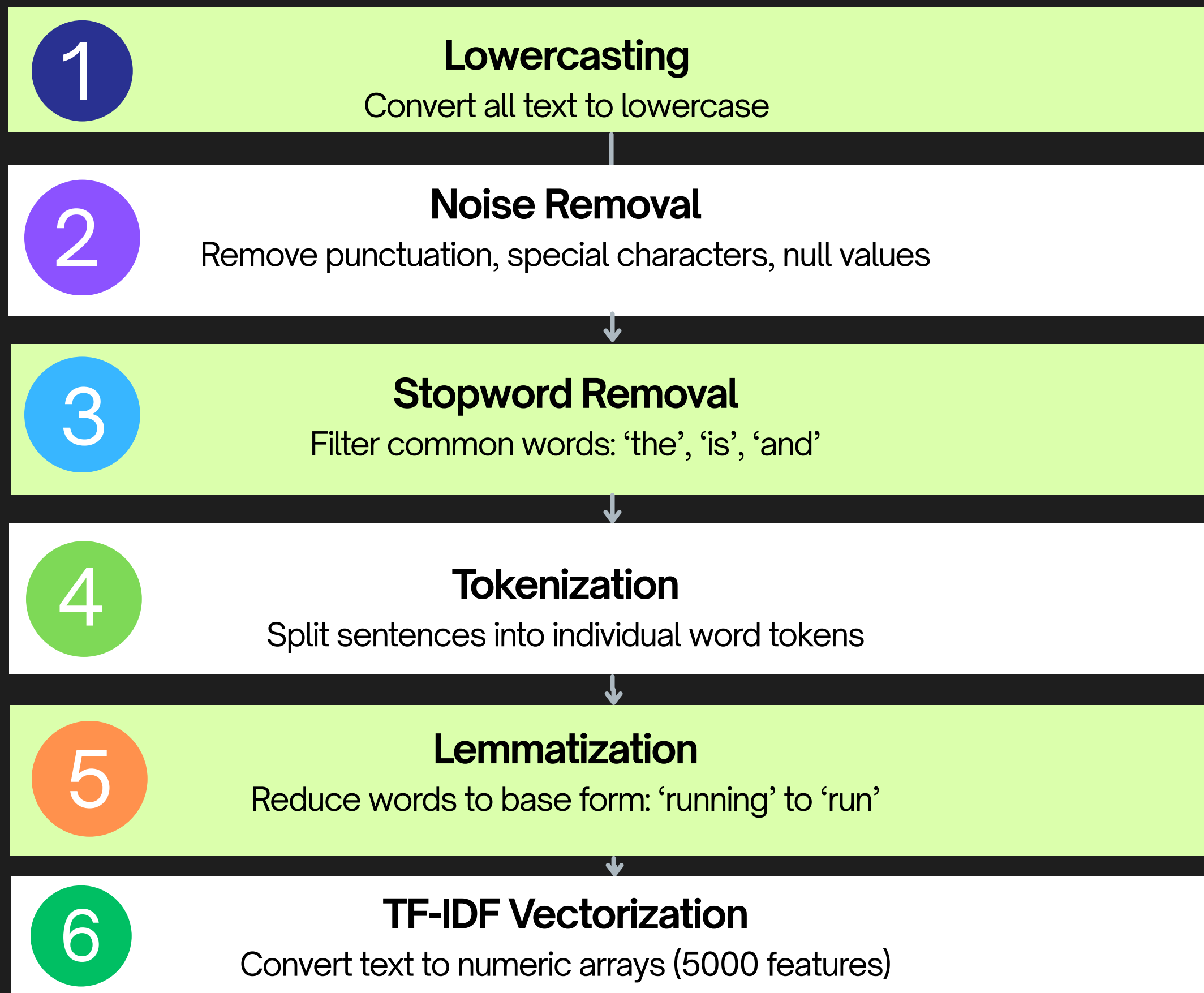


Merged Dataset

processed_fake_news_dataset.csv | 300+ MB
Binary Classification: Fake = 0 / Real = 1



NLP TEXT PREPROCESSING



Why TF - IDF ?

TF-IDF measures word importance relative to the corpus - giving higher weight to discriminating terms and lower weight to common ones

5000

max features selected

MACHINE LEARNING MODELS

Base Classifiers

K - Nearest Neighbors

Distance - based classification

Support Vector Machine

Hyperplane margin maximizer

Decision Tree

Rule - based tree classifier

Ensemble Learning

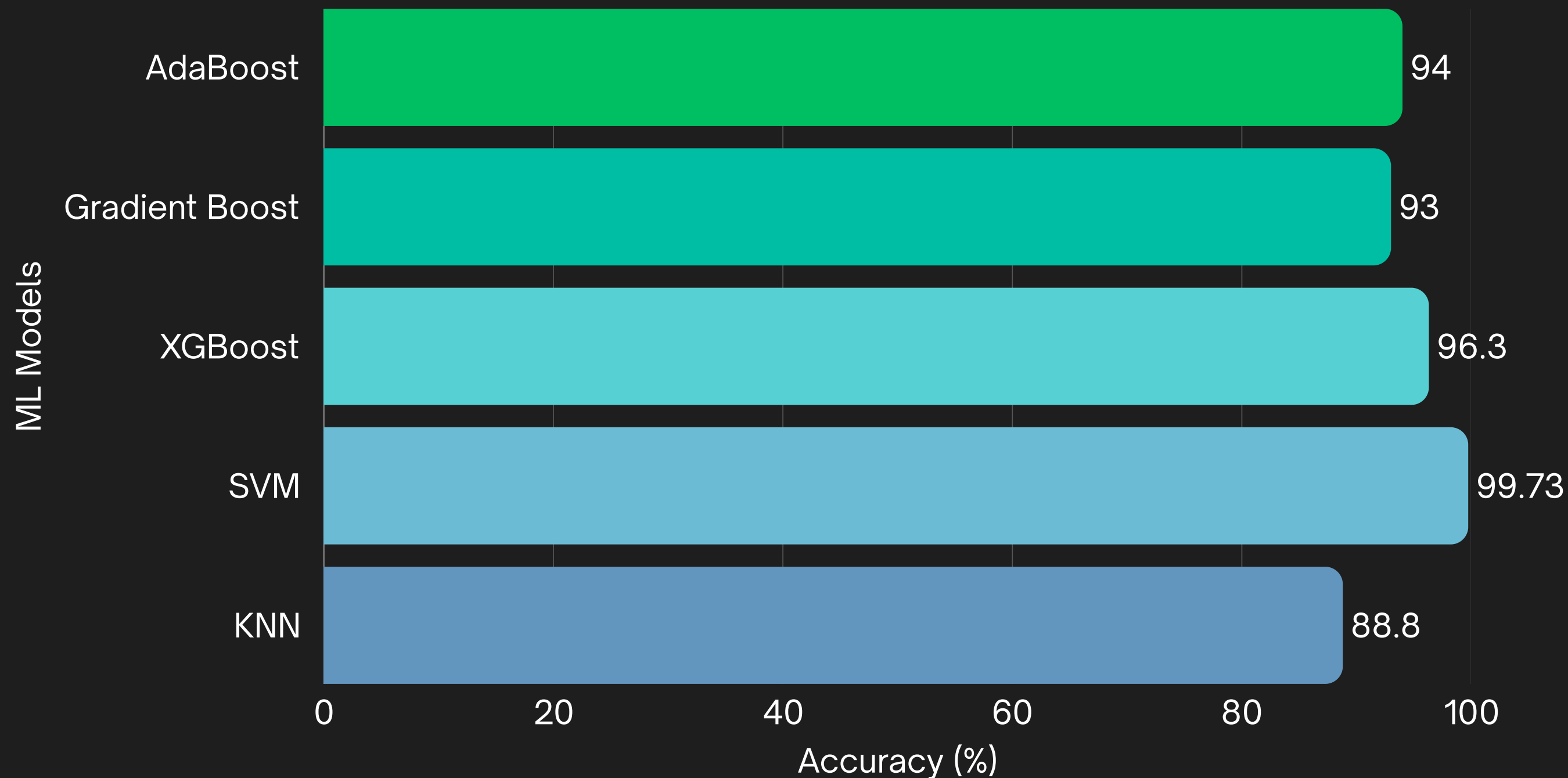
Adaboost

Adaptive boosting ensemble

XGBoost

Extreme gradient boosting

MODEL PERFORMANCE AND RESULT



~93.5
Adaboost & GBT

99.73%

SVM Accuracy

5 Models

Total Evaluated



Ensemble methods (AdaBoost, Gradient Boosting) achieved near-perfect accuracy by capturing complex non-linear textual relationships.

WEB DEPLOYMENT - STREAMLIT INTERFACE

1

Input

User pasics an unseen news article into the web interface

2

Preprocess

Real time NLP pipeline cleans and vectoriees the text

3

Predict

Pre-trained ensemble model analyces linguistic patterns

4

Output

Sysiem returns Fake / Real verdict with a confidence score

CONCLUSION AND FUTURE SCOPE

Conclusion

NLP combined with ensemble learning (AdaBoost, Gradient Boosting) delivers near-perfect accuracy proving that automated text classification is a viable and scalable solution for misinformation detection

~96.30%

Best Ensemble Accuracy

Future Scope



Live Data Integration

Scrape real time articles from Twitter/X and news portals



Deep Learning Models

Integrate deep learning models for contextual sentence understanding



Browser Extension

Verify news credibility in real time while browsing

THANK YOU